A Hierarchy of Risk Certificates for Forecasting Representations

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Abstract

A fitted forecast can fail an accuracy requirement even when its representation admits an adequate decoder. We study certificates that distinguish these possibilities for a fixed representation, metric, target, and decoder sensitivity budget. For bounded scalar targets, minimum squared errors on observation blocks, normalized by one less than block size, form a hierarchy whose first member is a pairwise obstruction. Expected scores are monotone and converge to optimal Lipschitz risk under unconditional iid sampling and compact metric support. Exact finite-block laws and a resolution analysis separate population relaxation, statistical calibration, and numerical slack. Jointly valid class-risk lower bounds and fitted-risk upper bounds can establish minimum feasible cost within a declared family. Reusing verified block duals provides an additional empirical-minimum calibration without further optimization solves. Controlled studies resolve necessity and adequacy; a natural-data census excludes one specified representation class. A complementary finite-cohort diagnosis certifies adequate capacity in 54 classes, including 39 for which both evaluated fitted heads fail. These results show why class inadequacy, fitted-model failure, and numerical prediction fidelity require different evidence. Learned KAHM representations provide a case study, with strong external comparators and unsuccessful screening and deployment results retained. Separate output checks certify numerical fidelity conditional on stored features, without implying future-response coverage or general forecasting superiority.

Keywords: risk certification, representation selection, Lipschitz regression, forecasting, KAHM

1 Introduction

Representation learning is usually assessed through the performance of fitted predictors. If a predictor misses a target accuracy, the observation alone does not show whether the representation is inadequate, the allowed decoder class is too restrictive, or the training procedure failed to access an adequate function. This distinction matters whenever a model must meet an error tolerance under a declared resource or sensitivity constraint.

We study a fixed observed-output prediction task. A representation Φ maps an available history X to coordinates U, and a declared metric and sensitivity budget specify the admissible decoders. A lower bound on the best risk in that class can rule out every admissible decoder. An upper bound on the risk of one admissible fitted predictor can establish adequacy. When every cheaper declared class is excluded and a fitted member of another class is accepted, the latter has minimum feasible cost within that family. A positive lower bound alone does not identify removable information loss: irreducible target noise and sensitivity restrictions also contribute.

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Object	Supported conclusion	Essential premise
Population class risk	Every decoder in a declared class exceeds a tolerance, or a fitted member meets it	Fixed class and loss; justified sampling; valid simultaneous calibration
Finite-cohort class risk	Exclusion or constructive adequacy on exactly the specified observations	Fixed coordinates, targets and class; sound optimization witnesses
Numerical reported output	Reported value lies within a numerical tolerance of a specified reference computation	Trusted feature input, fixed head semantics, valid arithmetic and identity checks

Table 1: The three uses of a certificate remain separate throughout the paper. No row implies the premises or conclusion of another.

The main construction strengthens a pairwise obstruction using observation blocks. Within each block we minimize squared error subject to all pairwise Lipschitz constraints, and normalize by one less than the block size. For scalar targets, the resulting expected scores increase with block size and approach optimal Lipschitz risk under the stated boundedness, compactness, and unconditional iid assumptions. This population result does not imply that larger blocks yield better finite-sample endpoints: each larger block consumes observations and reduces the number of independent scores available for calibration.

The paper separates three forms of assurance (Table 1). They share mathematical and computational tools but answer different questions. In particular, a numerical packet confirming a reported output is not a certificate that an unknown future response lies near that output.

Our contributions are organized around these questions.

1. **A centered scalar risk hierarchy.** We connect the exact pair score to larger constrained block minima, prove monotonicity and consistency in the stated setting, and characterize finite-block limitations. A resolution bound separates approximation bias, statistical uncertainty, and numerical slack.

2. **Computable and interpretable decisions.** Verified lower witnesses, independent calibration, fitted-risk upper bounds, and retained-block geometry support class exclusion, adequacy, and family-relative minimum-cost statements. Recycled block duals supply a second calibration without additional optimization solves. Standard concentration, scalar extension, and weak duality remain attributed foundations.

3. **Evidence about when the conclusions help.** Controlled decisions and a natural finite-cohort exclusion are complemented by a class-versus-head diagnosis, calibration limits, and unsuccessful screening and deployment studies. A separate numerical output check is evaluated on frozen forecasting policies with strong conventional controls.

KAHM dictionaries are an implementation case study rather than a premise of the risk theorem. Their reconstruction affinities can be retained and expanded while preserving earlier feasible decoders. Raw-history and Nyström representations supply essential controls. Scaling, local subspace cap, representation metric, and decoder budget are explicit parts of the candidate specification. The accumulated evidence does not establish general KAHM superiority or certificate-directed savings from avoiding model fits.

2 Related work and contribution boundary

Representations, KAHM and earlier prediction certificates. KAHM reconstruction models are established geometric feature constructions (Kumar et al., 2024, 2025). Their use here does not assume that reconstruction affinity implies predictive sufficiency. The earlier analysis of Kumar and Kargaran (2026) already provides prediction-risk certificates, fitted-risk upper bounds, empirical-minimum comparisons, and independent evaluation, including common-target forecasting comparisons. It also recognizes the need for joint coverage when lower and fitted upper certificates are combined. The present mathematical focus is the centered scalar observed-output hierarchy and decisions over a declared retained family. The comparison in Table 2 concerns central certificate constructions, not a claim that every ingredient in the present pipeline is new.

Sample-average optima and the centered hierarchy. Monotone lower expectations of ordinary sample-average optimal values are established in stochastic programming (Mak et al., 1999). Our statistic is $T_b = ba_b/(b-1)$, where a_b is the minimum empirical squared error under the fixed Lipschitz constraints. Since $b/(b-1)$ decreases, ordinary sample-average monotonicity alone does not imply monotonicity of this centered sequence. Translation of the fitted values and an average-over-subsets variance identity provide the required argument. The specific construction joins an exact positive-part pair member to a monotone scalar hierarchy, a compact-support population limit, and finite-block laws that expose its slack. None of these population statements implies that finite-sample confidence endpoints improve with block size.

Model-class falsification and empirical-infimum calibration. Lower inference for the best risk in a model class is an established problem (Müller et al., 2025). The empirical infimum is bounded above by the empirical risk of each fixed population competitor; this is the basis of earlier optimal-value lower confidence bounds (Guigues et al., 2017, Appendix A.2, equations (A10)–(A11)). The bounded-loss construction of Müller et al. (2025, Appendix A.1) also uses a nearly optimal fixed competitor. We specify the binary-KL inversion used here and prove how a pointwise optimization lower value can replace the exact empirical infimum. This is a calibration and computational construction for the declared problem, not a new generic concentration principle. Summing valid lower values from disjoint blocks supplies one such value; all observations, including those outside the blocks, retain their place in the full-sample normalization.

Pairwise obstructions and Lipschitz regression. The paired-residual argument in Akash et al. (2021, Lemma 2) uses target separation and a Lipschitz restriction to lower-bound an adversary's squared error. Finite-sample Lipschitz regression and extension also have established convex-optimization formulations (Gottlieb et al., 2017). We use these ingredients to study a conditional squared-positive-part witness, its exactness and limitations, and its relation to the centered block hierarchy. The conditioning law, marginal target risk, decoder budget and representation metric are declared together. A positive class-risk lower bound alone does not distinguish irreducible noise from representation loss or a restrictive sensitivity budget.

Calibrated decisions over representations. Learn then Test connects predictive risk control to simultaneous hypothesis testing (Angelopoulos et al., 2025); statistical selection among candidate state abstractions has earlier finite-sample guarantees in model-based reinforcement learning (Jiang et al., 2015). Here joint lower and upper endpoints answer a specific observed-output question: whether a declared decoder class is excluded by a loss tolerance, and whether a particular admissible

Question	Kumar and Kargaran (2026)	Present construction
Target of the central exclusion certificate	Successor soft-coordinate prediction error	Fixed observed-output squared risk
Decoder class for the central result	Linear coordinate predictors with a prescribed spectral-norm limit	Scalar Lipschitz class for the centered hierarchy; retained-block affine fits as a worked example
Lower-certificate construction	Class-centered variation and reference-label deviation; empirical-minimum comparator	Exact positive-part pair member and centered, normalized block hierarchy
Decision supplied	Class exclusion and fitted-risk upper bounds; joint coverage required for combined decisions	Centered-hierarchy and recycled-dual certificates; explicit minimum-feasible-cost conclusions within a declared family

Table 2: Comparison of the central certificate constructions with arXiv:2609.32652v3. The earlier work also includes fitted-risk upper bounds, independent evaluation and common-target forecasting comparisons. Standard concentration, empirical-infimum lower calibration and the KAHM reconstruction construction are shared foundations. The present hierarchy converges under the scalar, bounded-target, compact-support and unconditional iid assumptions stated in the theorem.

fitted predictor meets that same tolerance. Excluding all cheaper declared classes while accepting one fit establishes minimum feasible cost within that family. This decision can add necessity assurance even if upper-only selection chooses the same forecast. It does not imply a better forecast, a globally optimal representation, or a computational saving from certificate-directed search.

3 What a representation-risk certificate establishes

Let X denote the information available at a forecast origin and Y its scalar observed target. Condition on all development data and choices, so the representation $\Gamma : \mathcal{X} \to \mathcal{U}$, its measurable finite-valued pseudometric d, and a sensitivity budget $\kappa \geq 0$ are fixed. Assume Y lies almost surely in a known interval $I = [y_-, y_+]$ of diameter $D > 0$, and put $B = D^2$. Define

$$\mathcal{G}_\kappa(d) = \{g : \mathcal{U} \to I \text{ measurable} : |g(u) - g(v)| \leq \kappa d(u, v) \text{ for all } u, v\}, \tag{1}$$

$$R(g) = \mathbb{E}[(Y - g(\Gamma(X)))^2], \qquad R^*_\kappa = \inf_{g \in \mathcal{G}_\kappa(d)} R(g). \tag{2}$$

All displayed infima and statistics are assumed measurable. The expectation uses the declared evaluation law, conditional on development, and not the law conditional on an individual query. Scalar clipping to I preserves the Lipschitz constraint and decreases squared loss. Hence this bounded class has the same optimal risk as the corresponding unrestricted scalar Lipschitz class. Constants belong to it, giving $0 \leq R^*_\kappa \leq \mathrm{Var}(Y) \leq B/4$. When $D = 0$, the target and interval-valued predictions are the same constant; all risks and certificates are zero, without any division by B.

The representation, metric, budget, target, loss, and evaluation law jointly specify the certification problem. Changing feature scaling while retaining the same numerical budget generally changes the class. A ridge penalty does not establish a hard Lipschitz budget. A fitted upper endpoint can be

paired with a class lower endpoint only after establishing that the fitted decoder $\widehat{g}$ belongs to that class.

The standard conditional-expectation decomposition explains the scope of this risk. With $M_X = \mathbb{E}[Y \mid X]$ and $M_U = \mathbb{E}[Y \mid \Gamma(X)]$, orthogonality in L^2 gives

$$R^*_\kappa = \underbrace{\mathbb{E}(Y - M_X)^2}_{\text{future uncertainty}} + \underbrace{\mathbb{E}(M_X - M_U)^2}_{\text{conditional-mean information loss}} + \underbrace{\inf_{g \in \mathcal{G}_\kappa(d)} \mathbb{E}(M_U - g(\Gamma(X)))^2}_{\text{sensitivity restriction}}. \tag{3}$$

The three residual components are pairwise orthogonal because $\Gamma(X)$ is measurable with respect to X; taking the infimum yields the identity. A positive lower risk certificate does not separate these terms. It is therefore neither a certificate of unrestricted information loss nor a sufficient rule for discovering a successful refinement.

3.1 The pairwise starting point

Draw a variable A, and then draw (U, Y) and (U', Y') independently from the same conditional law given A, where each copy has the declared marginal law and $U = \Gamma(X)$. With $z_+ = \max\{z, 0\}$, define

$$W_{\kappa,A} = \frac{1}{2}\mathbb{E}\big(|Y - Y'| - \kappa d(U, U')\big)_+^2. \tag{4}$$

Taking A constant gives unconditional independent pairing. A common forecast origin or a frozen stratum gives a different admissible construction when its conditional independence and marginal-law conditions hold. Arbitrary nearest-neighbor matching need not meet these conditions.

Proposition 1 (Pairwise obstruction) *Under this pairing construction, $W_{\kappa,A} \le R^*_\kappa$.*

Proof For any fixed admissible g, put $r = Y - g(U)$ and $r' = Y' - g(U')$. The triangle inequality implies $(|Y - Y'| - \kappa d(U, U'))_+ \le |r - r'|$. Conditional independence gives $\frac{1}{2}\mathbb{E}(r - r')^2 = \mathbb{E}r^2 - \mathbb{E}\{\mathbb{E}[r \mid A]^2\} \le R(g)$. Square the pointwise inequality, take expectations, and then take the infimum over g. ∎

Paired Lipschitz obstruction arguments have prior precedents (Akash et al., 2021). The role of the pair score here is its exact connection to a centered hierarchy, including explicit cases in which pairwise information is arbitrarily loose.

4 A centered hierarchy and its finite-block limits

For $b \ge 2$ unconditional iid observations (U_i, Y_i) from the fixed evaluation law, define

$$T_b = \frac{1}{b-1} \min_{v \in \mathbb{R}^b} \left\{ \sum_{i=1}^{b} (Y_i - v_i)^2 : |v_i - v_j| \le \kappa d(U_i, U_j) \quad (i, j \le b) \right\}, \qquad \mathcal{W}_b = \mathbb{E}T_b. \tag{5}$$

The feasible set is nonempty and closed and the objective is strictly convex and coercive, so the minimizer exists and is unique. Clipping its values to I preserves feasibility and cannot increase the objective. If a_b is the usual empirical minimum MSE, then $T_b = b a_b/(b-1)$. The factor $1/(b-1)$ exploits translation-invariant difference constraints; it is not a generic normalization for empirical risk minimization.

Theorem 2 (Centered scalar block hierarchy) *For the fixed class in* (1) *and bounded scalar target above:*

1. $T_2 = \frac{1}{2}(|Y_1 - Y_2| - \kappa d(U_1, U_2))_+^2$, *so* $\mathcal{W}_2 = W_{\kappa,A}$ *for constant* A.
2. *With*
$$C_b = \frac{B\lfloor b^2/4 \rfloor}{b(b-1)}, \qquad 0 \le T_b \le C_b, \qquad \mathcal{W}_b \le R_\kappa^*. \tag{6}$$
3. *If* $T_m(S)$ *denotes the score on a subset* S *of* m *observations from a realized* b*-block, then for* $2 \le m \le b$*,*
$$\binom{b}{m}^{-1} \sum_{|S|=m} T_m(S) \le T_b. \tag{7}$$
Consequently $\mathcal{W}_2 \le \mathcal{W}_3 \le \cdots \le R_\kappa^*$.
4. *If* $(\mathcal{U}, d)$ *is a compact metric space, then* $\mathcal{W}_b \uparrow R_\kappa^*$ *as* $b \to \infty$.

Proof For two observations, if the target separation exceeds the allowed fitted separation by $e > 0$, the optimum has opposite residuals $e/2$ and $-e/2$. Its squared residual sum is $e^2/2$. Otherwise both targets are feasible. This proves the first assertion.

For any fixed admissible g, let $r_i = Y_i - g(U_i)$ and $\bar{r} = b^{-1}\sum_i r_i$. The translated fitted values $g(U_i) + \bar{r}$ satisfy every difference constraint. Thus

$$T_b \le \frac{1}{b-1}\sum_i (r_i - \bar{r})^2, \qquad \mathcal{W}_b \le \mathrm{Var}(Y - g(U)) \le R(g).$$

Taking the infimum gives the risk bound. Choosing the constant fitted value $\bar{Y}$ bounds T_b by the unbiased sample variance of $Y_1, \ldots, Y_b$. That variance is convex in each target coordinate; its maximum on I^b occurs at interval endpoints. With k upper endpoints its value is $Bk(b-k)/\{b(b-1)\}$. Maximizing over integer k proves the range.

Let v minimize the full-block objective and set $r_i = Y_i - v_i$. Translation invariance implies $\sum_i r_i = 0$. Therefore

$$T_b = \frac{1}{b(b-1)}\sum_{i<j}(r_i - r_j)^2.$$

Restricting v to S and translating by that subset's residual mean gives

$$T_m(S) \le \frac{1}{m(m-1)}\sum_{i<j,\ i,j\in S}(r_i - r_j)^2.$$

A uniformly chosen m-subset contains each pair with probability $m(m-1)/\{b(b-1)\}$. Averaging yields (7); taking expectations gives monotonicity. This does not order arbitrary realized blocks of different sizes.

For consistency, every finite feasible vector admits the scalar extension $g_v(u) = \min_i\{v_i + \kappa d(u, U_i)\}$. The triangle inequality makes it κ-Lipschitz and the finite constraints give $g_v(U_i) = v_i$. After clipping to I, this identifies a_b with the empirical infimum over the same bounded Lipschitz class as the population problem. On a compact metric space that class is compact in the uniform

norm by boundedness and equicontinuity. Choose a finite uniform $1/j$-net for each integer j. The strong law holds simultaneously on their countable union. Since

$$|(y - f(u))^2 - (y - g(u))^2| \le 2D\|f - g\|_\infty \quad (y, f(u), g(u) \in I),$$

net approximation yields uniform convergence of empirical risks and hence $a_b \to R^*_\kappa$ almost surely. Now $T_b = ba_b/(b-1)$ and $0 \le T_b \le B/2$. Dominated convergence proves the claimed limit. ■

The strengthened information comes from requiring jointly consistent fitted values across all pairs in a block. Monotonic expected sample-average minima and batch lower estimation are established precedents (Mak et al., 1999; Linderoth et al., 2006). The specific connection here is the centered normalization, its exact pair member, its subset inequality, and its Lipschitz-risk limit. It does not supply a new concentration inequality or a vector-target consistency theorem.

4.1 No fixed block size is uniformly informative

Proposition 3 (Rare-conflict and noisy-support laws) *For $(U, Y) = (3, 0), (0, -1), (1, 1)$ with probabilities $1 - 2\varepsilon, \varepsilon, \varepsilon$, where $0 < \varepsilon < 1/2$, and $\kappa = 1$ with ordinary distance,*

$$R^*_\kappa = \varepsilon/2, \qquad \mathcal{W}_b = \frac{2b\varepsilon - 1 + (1 - 2\varepsilon)^b}{4(b-1)}. \tag{8}$$

For U uniform on M points with pairwise distances at least D/κ, where $D, \kappa > 0$, and an independent equiprobable target $Y \in \{0, D\}$,

$$R^*_\kappa = D^2/4, \qquad \mathcal{W}_b = \frac{D^2}{4(b-1)} \left[b - M\{1 - (1 - 1/M)^b\} \right]. \tag{9}$$

For each fixed b, the ratio $\mathcal{W}_b/R^*_\kappa$ tends to zero as $\varepsilon \downarrow 0$ in the first law or $M \to \infty$ in the second.

Proof In the first law, the two rare fitted values must differ by at most one. Their minimum unweighted sum of squared errors is $1/2$, attained at $-1/2$ and $1/2$; the common atom can be fitted at zero. A scalar Lipschitz extension realizes these values, proving the population optimum. In a block, let N_-, N_+ count the rare atoms and put $N = N_- + N_+$. Their constrained weighted residual sum is N_-N_+/N when $N > 0$ and zero otherwise. Since $N \sim \text{Binomial}(b, 2\varepsilon)$ and $N_- \mid N \sim \text{Binomial}(N, 1/2)$, the expected sum is $\{\mathbb{E}N - \Pr(N \ge 1)\}/4$. Dividing by $b - 1$ proves (8).

In the second law, all interval-valued fitted values at distinct support points satisfy the distance constraints. At a point observed $n_j > 0$ times, fitting its target mean has expected residual sum $(n_j - 1)D^2/4$. If O_b is the number of occupied points, the total expected sum is $D^2\mathbb{E}(b - O_b)/4$, and $\mathbb{E}O_b = M\{1 - (1 - 1/M)^b\}$. The constant $D/2$ attains the population risk $D^2/4$ because the target is independent of U. The stated limits follow from the displayed formulas. ■

The noisy-support examples are finite compact spaces; their diameter can remain D/κ as M grows. Thus block size and target range, even together with a bound on representation diameter, do not yield a uniform vanishing approximation error over these spaces. Consistency is a statement for a fixed evaluation law, not a universal finite-block guarantee.

Common forecast origins require a different limit. If the observations in a block are independent only conditional on a common A, validity and monotonicity still hold. For finite-valued A under the corresponding compactness conditions, the limiting quantity is

$$\sum_a \Pr(A = a) \inf_{g \in \mathcal{G}_\kappa(d)} \mathbb{E}[(Y - g(U))^2 \mid A = a].$$

It need not equal the risk of one shared optimal decoder. For example, with constant U and a target determined by a nonconstant common origin, all within-origin block scores vanish although the shared constant decoder has positive risk. Drawing an independent origin in each independent episode instead produces unconditional iid observations from the declared episode–origin mixture.

5 Computable lower values and simultaneous calibration

Optimization validity and statistical calibration are separate steps. A primal objective bounds an empirical minimum from above and therefore cannot supply the lower value needed below. Sparse or unfinished dual calculations can supply valid lower values if their arithmetic is verified.

5.1 A deterministic finite-cohort certificate

For any fixed cohort $(u_i, y_i)_{i=1}^n$ with $y_i \in I$, define $a_n = \inf_{g \in \mathcal{G}_\kappa(d)} n^{-1} \sum_i (y_i - g(u_i))^2$. The scalar extension used above identifies na_n with the finite quadratic minimum under all pairwise Lipschitz constraints. This equality uses no independence or population sampling assumption.

Proposition 4 (Verified graph duals and disjoint reuse) *Orient any selected graph of pair constraints and let M be its incidence matrix, with one $+1$ and one -1 in each column. For a finite signed multiplier vector λ, put $s = M\lambda$. Then*

$$D_G(\lambda) = s^\top y - \kappa \sum_{e=(i,j)} d(u_i, u_j)|\lambda_e| - \frac{1}{4}\|s\|_2^2 \;\le\; na_n. \tag{10}$$

Let $S_1, \dots, S_K$ be disjoint subsets of cohort indices and let $\underline{A}_k$ be verified nonnegative lower bounds on their minimum squared residual sums. Then

$$d_n = \frac{1}{n} \sum_k \underline{A}_k \le a_n. \tag{11}$$

Uncovered observations remain in the denominator n.

Proof For a feasible vector v, the term $\lambda^\top M^\top v - \kappa \sum_e d(u_i, u_j)|\lambda_e|$ is nonpositive. Add it to $\|y - v\|_2^2$ and minimize over all v. The minimizer is $y - s/2$, with value (10), proving weak duality without a convergence or optimality assumption. For (11), restrict any full feasible vector to each disjoint subset. Its residual sum on that subset is at least the corresponding minimum; residuals at uncovered observations are nonnegative. Summing and taking the full infimum proves the result. ■

For equal blocks of size b and $\underline{T}_k = \underline{A}_k/(b-1)$, the reused value is

$$d_n = \frac{K(b-1)}{n} \left(\frac{1}{K} \sum_k \underline{T}_k \right). \tag{12}$$

This factor separates the centered block statistic from a full empirical MSE. On disjoint vertex sets the graph-dual objectives add exactly. Objectives from overlapping blocks cannot simply be summed because their squared-norm terms have cross terms. They may instead be embedded in one graph whose full objective is verified, or combined by a maximum of lower values for the same a_n. Reuse requires no additional optimization solve; it still incurs arithmetic and calibration work.

5.2 Two lower calibrations with different independent sample sizes

For $q, p \in [0,1]$, let

$$\mathrm{kl}(q\|p) = q\log(q/p) + (1-q)\log\{(1-q)/(1-p)\},$$

with the usual zero-numerator and infinite boundary conventions. For $c > 0$, let $k_-(q,c)$ and $k_+(q,c)$ be the smallest and largest p with $\mathrm{kl}(q\|p) \le c$. Both are nondecreasing in q; increasing c lowers k_- and raises k_+. The established bounded-mean Chernoff argument gives, for iid $V_i \in [0,C]$, each of the one-sided statements

$$Ck_-(\bar{V}/C, \log(1/\alpha)/N) \le \mathbb{E}V_1, \qquad \mathbb{E}V_1 \le Ck_+(\bar{V}/C, \log(1/\alpha)/N) \tag{13}$$

with failure probability at most α (Hoeffding, 1963). The observations need not be Bernoulli. Indeed, convexity bounds their exponential moments by $1 - p + pe^t$, with $p = \mathbb{E}V_1/C$. Independence and exponential Markov yield $\exp\{-N\,\mathrm{kl}(q\|p)\}$ for the appropriate one-sided empirical tail; inversion gives (13). The two directions require separate allocation when both are claimed. At $C = 0$ use zero endpoints.

Proposition 5 (An empirical-infimum lower endpoint) *Let a fixed nonempty class have losses in $[0,B]$ and measurable empirical infimum a_n on n iid observations. Any measurable returned value $0 \le d_n \le a_n$ pointwise satisfies*

$$Bk_-(d_n/B, \log(1/\alpha)/n) \le \inf_g \mathbb{E}\ell_g \tag{14}$$

with probability at least $1-\alpha$.

Proof Choose fixed population competitors g_j whose risks r_j decrease to $r^* = \inf_g \mathbb{E}\ell_g$. They are proof devices, not predictors chosen from calibration. Since $d_n \le a_n \le n^{-1}\sum_i \ell_{g_j}(X_i,Y_i)$, monotonicity of k_- and (13) give $\Pr\{Bk_-(d_n/B,c) > r_j\} \le \alpha$ for each j. These events increase to $\{Bk_-(d_n/B,c) > r^*\}$, where $c = \log(1/\alpha)/n$. Continuity of probability proves the claim even when the infimum is not attained. ■

This fixed-competitor argument needs no uniform-complexity penalty for a class *lower* endpoint. It supplies neither a fitted-on-calibration upper bound nor an excess-risk guarantee. It is an inherited calibration ingredient, not a new general principle of statistical class falsification.

Now freeze a finite family of classes m and fitted members $\widehat{g}_m$ using independent development data. Each record uses its declared known range B, ordinary squared loss, target, and evaluation law. From n iid calibration observations, form $K_b = \lfloor n/b \rfloor \ge 1$ predetermined disjoint blocks and

pointwise lower scores $0 \le \underline{T}_{m,b,k} \le T_{m,b,k}$. Also allow any pointwise value $0 \le d_{m,n} \le a_{m,n}$ and verified fitted-loss upper values $\ell_{m,i} \le \overline{\ell}_{m,i} \le B$. For separately allocated probabilities, set

$$L_{m,b}^{\mathrm{blk}} = C_b k_- \left(\frac{K_b^{-1} \sum_k \underline{T}_{m,b,k}}{C_b}, \frac{\log(1/\alpha_{m,b}^{\mathrm{blk}})}{K_b} \right), \tag{15}$$

$$L_m^{\mathrm{emp}} = B k_-(d_{m,n}/B, \log(1/\alpha_m^{\mathrm{emp}})/n), \tag{16}$$

$$U_m = B k_+ \left(\frac{n^{-1} \sum_i \overline{\ell}_{m,i}}{B}, \frac{\log(1/\alpha_m^{\mathrm{up}})}{n} \right). \tag{17}$$

Take L_m to be the maximum of the covered lower endpoints, or zero when none are used. Numerical implementations report lower endpoints no larger and upper endpoints no smaller than these exact-real quantities.

Theorem 6 (Simultaneous class and fitted-risk certificate) *If the finite event ledger is fixed before calibration and its allocated failure probabilities sum to at most δ, then, conditional on development, with probability at least $1 - \delta$,*

$$L_m \le R_m^* \le R_m(\widehat{g}_m) \le U_m \tag{18}$$

simultaneously for every covered record. Each block endpoint separately also lower-bounds its $\mathcal{W}_{m,b}$. Different events may reuse data.

Proof For each block event, apply the lower inequality in (13) to the exact iid scores $T_{m,b,k} \in [0, C_b]$. Their verified lower mean cannot increase that endpoint. Theorem 2 then gives the class lower bound. Proposition 5 gives the empirical-infimum event. For a fitted upper, apply the upper inequality in (13) to the exact iid losses; a larger verified mean is conservative. Membership supplies $R_m^* \le R_m(\widehat{g}_m)$. A conditional union bound gives the common event; maxima of its lower endpoints and outward numerical arithmetic preserve it. Integrating over development yields the same unconditional probability statement. ■

The block endpoint calibrates K_b independent scores, not n individual observations. Larger blocks strengthen $\mathcal{W}_b$ but reduce K_b; their finite-sample endpoints need not improve. The empirical-infimum endpoint instead uses n for the distinct pointwise objective a_n. Neither calibration uniformly dominates the other. Numerical paths, graphs, or partitions may be chosen adaptively when they still return a pointwise lower value for the same fixed a_n. This does not authorize changing a prediction class after inspecting calibration. Additional block sizes, class choices, or sample-size looks require their own simultaneous or sequential validity argument.

Independent episodes may contain arbitrarily dependent windows. An episode average is one bounded fitted-loss observation. One independently sampled origin per episode supplies the unconditional iid scalar observations for (5). Counting overlapping windows as independent calibration units is not justified by this theorem. Training extrema do not establish an almost-sure target range, and capped-loss upper bounds cannot be combined with the ordinary-MSE lower construction.

5.3 Arithmetic semantics and the scope of a numerical certificate

For (10), treat retained multipliers and parameters according to their declared exact semantics. An upper enclosure of each distance, a lower enclosure of the linear term, and upper enclosures of

subtracted terms give a valid lower objective. The lower KL root and its final rescaling must be rounded down; the upper root and rescaling must be rounded up. Both directions use an upper enclosure of $\log(1/\alpha)/N$. Rigorous deterministic arithmetic consumes no extra statistical failure probability.

The declared feature map and the evaluated head must match the certificate. A guarantee conditional on stored feature vectors does not certify their upstream computation. Likewise, a mathematically Lipschitz affine head and a rounded prediction program are distinct objects: rounding can introduce discontinuities. Verifying output fidelity to a fixed head does not establish that head's future prediction risk. These are separate statistical and numerical claims. Coupled numerical enclosures transfer safely through the monotone KL maps above; a variance-dependent empirical Bernstein endpoint additionally needs a valid variance enclosure or iid computed scores.

6 Refinement and decisions within a declared family

Let Φ_j be fixed feature blocks and retain $\Gamma_s = (\Phi_1, \dots, \Phi_s)$ with metric $\rho_s(u, v) = \max_{j \le s} \|u_j - v_j\|_2$. The scalar clipped affine family

$$\mathcal{F}_{s,\kappa} = \left\{ x \mapsto \mathrm{clip}_I \left(a + \sum_{j \le s} \theta_j^\top \Phi_j(x) \right) : \sum_{j \le s} \|\theta_j\|_2 \le \kappa \right\} \tag{19}$$

is a fitted subclass of the Lipschitz comparator. Coefficients and budgets must use the physical target units after any inverse target transform.

Proposition 7 (Retained blocks preserve the comparator) *Every member of* (19) *is κ-Lipschitz in ρ_s. Appending an unchanged old-coordinate stack and a new block embeds the earlier feasible family and cannot increase its optimal risk. Appending an exact duplicate block leaves that predictor class and its induced distance unchanged. The corresponding inclusions and duplication invariance also hold for all Lipschitz predictors on the retained stack.*

Proof Cauchy–Schwarz and the triangle inequality bound the unprojected prediction difference by $\sum_j \|\theta_j\|_2 \|u_j - v_j\|_2 \le \kappa \rho_s(u, v)$. Clipping is nonexpansive. Assigning zero coefficients to a new block embeds every old predictor. For a duplicate, merge its coefficients with those of the copied block; the triangle inequality shows that the total norm does not increase. This gives reverse inclusion. The maximum distance is unchanged by an exact duplicate. For a general Lipschitz decoder, composition with the coordinate projection embeds the old class because that projection is nonexpansive. For duplication, composition with $u \mapsto (u, u_j)$ gives the reverse inclusion because this map preserves the maximum metric. ■

Group norm constraints are established tools (Bach, 2008); here they keep the sensitivity meaning compatible with retained blocks. These are class-inclusion statements, not monotonicity claims for separately trained estimators. A squared ridge penalty can change under duplication even when the feasible class does not. Refitting old blocks or changing the budget is a new comparison, not this particular nesting construction.

Corollary 8 (Adequacy, exclusion, and minimum feasible cost) *Fix an MSE tolerance $\tau \geq 0$ and class costs $c_m \geq 0$ before calibration. On the joint event* (18)*, $U_m \leq \tau$ certifies an exhibited adequate predictor, while $L_m > \tau$ excludes every decoder in that class. Otherwise the decision is unresolved. Define a feasible class by the existence of a member with risk at most τ, and let c^* be the minimum cost of a feasible class. With the minimum of an empty set taken as $+\infty$,*

$$c_- := \min_{m:L_m \leq \tau} c_m \;\leq\; c^* \;\leq\; c_+ := \min_{m:U_m \leq \tau} c_m. \tag{20}$$

If $c_- = c_+ < \infty$, an accepted fit attains the minimum feasible cost within the declared family. If every class is excluded, that family is infeasible at the requested tolerance.

Proof Every accepted fitted member exhibits feasibility. Every genuinely feasible class is nonexcluded, since its lower endpoint cannot exceed the risk of an adequate member. These inclusions give the cost interval and both conclusions without assuming attainment of a population infimum. ■

An RMSE tolerance ϵ uses $\tau = \epsilon^2$. With several required horizons, acceptance requires all covered upper tests, while one covered lower test suffices to exclude a compatible joint class. When $c_- < c_+$, the selected fit has minimum cost only among accepted fits. Feature count, stored bytes, and runtime are different cost definitions; the cost bracket does not establish savings in the total resources used to search, fit, and certify candidates.

The comparison $L_c - U_r > 0$ certifies that the exhibited replacement outperforms every member of the coarse class by at least $L_c - U_r$. Calling this a retained-coordinate refinement additionally requires the class inclusion above. Its successful discovery is an empirical question. Before attempting any adequacy test, the KL upper endpoint has the deterministic zero-loss floor

$$U \geq B(1 - \alpha^{1/n}). \tag{21}$$

This follows from monotonicity and $k_+(0, c) = 1 - e^{-c}$. It is a necessary feasibility check, not a power guarantee for a nonzero-risk predictor. A certificate can be valid yet too weak to decide the requested tolerance.

Resolution beyond validity. Appendix A bounds finite-block population bias using an internal uniform cover and gives a sufficient decision-resolution inequality. It separates that bias from statistical calibration and verified numerical slack. These sufficient bounds explain why a valid certificate can remain inconclusive; they do not claim an optimal sample-complexity rate.

7 Fixed representations and forecasting procedures

The history length, forecast horizon, observed target, and origin rule define the task before representations are compared. Development comprises feature fitting, head fitting, and permitted model selection. Population calibration uses independent units after the candidate family is frozen. Chronological assessment on a single observed series has a different evidential role and is reported as a finite-period comparison.

7.1 KAHM as an expandable example

KAHMs use local kernel reconstruction models fitted on training histories (Kumar et al., 2024, 2025; Kumar and Kargaran, 2026). For a preprocessed history $z(x)$ and reconstruction $\widehat{z}_{bcj}(x)$ in reference

group c of retained feature block b, write

$$e_{bcj}(x) = 1 - \exp\{-\|z(x) - \widehat{z}_{bcj}(x)\|_2\}, \qquad \theta_{bcj}(x) = \pi^{-1} \arccos \gamma_{bcj}(x).$$

Here γ is clipped cosine similarity, with the declared zero-norm convention. The group discrepancy and normalized association are

$$T_{bc}(x) = \min_j \sqrt{\{e_{bcj}(x)^2 + \theta_{bcj}(x)^2\}/2},$$
$$\Phi_{bc}(x) = \frac{(1 - T_{bc}(x) + \tau)^\omega}{\sum_r (1 - T_{br}(x) + \tau)^\omega}, \qquad \tau > 0,\ \omega > 0.$$

These simplex coordinates are reconstruction affinities, not calibrated response probabilities. Stable log-space normalization is part of the implementation. The local PCA request is $r_0 = \min(r_{\max}, d_x, N_{\mathrm{loc}} - 1)$, followed by the inherited variance and projected-range rules; requested cap and stored rank are different quantities.

Appending fixed coordinate blocks produces $\Gamma_s = (\Phi_1, \ldots, \Phi_s)$. Under the maximum of within-block Euclidean distances, the product of simplices is compact with diameter at most $\sqrt{2}$, independently of the number of blocks. Thus the scalar hierarchy applies to any bounded scalar target under unconditional iid sampling, even if the history-to-feature map is discontinuous. Adding feature blocks changes the candidate class; increasing the number of observations in a certificate block changes the relaxation for a fixed class.

Training-only min–max and standard scaling are both retained where declared. Scaling can change the radial and angular terms, learned partitions, and local ranks. It must not be treated as an innocuous coordinate change while keeping the same numerical sensitivity budget. For example, replacing U by cU preserves a Lipschitz class only after the metric or budget is compensated. Likewise, for fixed positive scores, changing the concentration ω is an invertible reparameterization in exact arithmetic, yet can alter accessibility to a restricted decoder and numerical conditioning.

7.2 Theoretical family and implemented heads

A retained-block affine head with $\sum_b \|W_b\|_{\mathrm{F}} \le \kappa$ is κ-Lipschitz in the maximum-block metric. Projection onto a known output box is nonexpansive. Zero padding retains earlier feasible forecasts, while merging coefficients of identical blocks proves invariance to exact block duplication. These are statements about feasible classes. A ridge penalty can change under duplication, and a fitted predictor need not improve when its class grows.

The independent-episode controlled study uses this retained-block family. The later chronological studies instead declare fixed Euclidean feature metrics and their own coefficient constraints. Their finite-cohort results are not silently transferred to the retained-block population theorem. The final affine protocol uses a rational head, a Euclidean norm bound of two, exact intercept recentering, and normalized output clipping. Its training-derived output range defines that predictor; it is not assumed to contain every future target.

Costs must also be specified. Coordinate count can order an abstract family, but it does not establish an ordering of complete storage or runtime. When deployment utility is assessed, feature extraction, stored encoder state, head evaluation, checking, and any cached state must be included or explicitly separated.

8 Empirical evaluation

The evaluation examines controlled population-risk decisions, natural-data class exclusion, finite-cohort class-versus-head diagnosis, and predictive and operational performance. These studies have distinct protocols and evaluation laws; their confidence allocations apply within each declared study.

8.1 Controlled necessity, adequacy, and calibration limits

In the controlled study, each history coordinate is $X_j = S_j(0.9 + V_j)$, with independent equiprobable signs and $V_j \sim \mathrm{Unif}[-0.1, 0.1]$. The repairable target is $Y = 0.8X_1X_2 + E$, where $E \sim \mathrm{Unif}[-0.05, 0.05]$. The irreducible control replaces X in the response by an independent unavailable history with the same law. Both targets and clipped fitted outputs lie in $[-0.85, 0.85]$, so the squared-loss range is $B = 2.89$.

Twelve complete learning repetitions per regime select a geometry shared between their two prefixes before independent calibration, preserving two- and six-coordinate nested prefixes. The sensitivity budget is $\kappa = 2$ and the design tolerance is MSE 0.1. Both calibrations reuse the same block optimization outputs and the same fitted forecasts. At the prespecified primary size $n = 1024$, both resolve all twelve repetitions in each regime (Table 3). In the repairable regime, the coarse class is excluded and the refined fitted head is accepted; in the independent-future regime, both classes are excluded. The conclusion applies to the selected geometry, not every geometry considered during development.

Regime	Calibration size	Both resolve	Block only	Recycled only	Neither
Repairable	256 (secondary)	0	0	0	12
Repairable	1024 (primary)	12	0	0	0
Irreducible	256 (secondary)	1	0	11	0
Irreducible	1024 (primary)	12	0	0	0

Table 3: Controlled decision counts with matched optimization outputs. Both methods share representations, observations, block programs and fitted upper endpoints. Recycled calibration adds no optimization solve; it still has arithmetic and bookkeeping costs.

The simultaneous allocation is $\alpha = 0.05/1008$ per designated endpoint. At $n = 256$, even a zero-loss fitted head has KL upper endpoint

$$B(1 - \alpha^{1/n}) = 0.109752706592 > 0.1.$$

Refinement acceptance is therefore impossible at that size under the declared formula, regardless of how sharply the lower risk is optimized. At least 282 independent observations are necessary even for the zero-loss extreme. This is a formula-specific resolution limit, not a universal sample-complexity lower bound.

At $n = 256$, recycled calibration resolves eleven additional irreducible-regime repetitions by excluding both classes, but it adds no primary decisions. Neither improved lower endpoints nor their reuse changes the terminal predictions. The experiment uses guarded numerical replay and common feature sentinels; it is not an end-to-end interval proof of native feature construction.

8.2 Natural-data exclusion of a specified class

The natural-data exclusion study uses a fixed cohort of 8192 PTB-XL patients (Wagner et al., 2020, 2022). The bounded task predicts a clipped, normalized lead-II amplitude 40 ms ahead from 32 past measurements spanning 310 ms. The selected two-coordinate map uses min–max preprocessing, local cap two, concentration 0.5, and maximum-block Euclidean sensitivity budget $\kappa = 1$.

Exact rational evaluation of retained dual witnesses, conditional on the inherited canonical-feature enclosures, gives

$$R^{\star}_{\mathrm{cohort},\,2,\,1} \;\geq\; 0.00643206 \;>\; 0.0058458874,$$

where the lower display is rounded downward and the right-hand value approximates the unchanged conservative requirement threshold. The retained threshold is an upper enclosure of 1.1 times the frozen reference's cohort MSE; exceeding it proves failure of the unchanged 10% allowance. Every decoder in this specified class fails that requirement. This is stronger than the failure of its fitted head, but it concerns the census rather than a future-patient population.

The six- and fourteen-coordinate lower endpoints are 0.00515356 and 0.00495880, rounded downward, and do not exclude those classes. Their separately selected geometries are not a nested feature hierarchy. A retained witness additionally gives the necessary budget $\kappa \geq 54.82579$ for adequacy of the same two-coordinate geometry, rounded downward. It is not sufficient and is coordinate dependent. The evaluated KAHM predictors did not satisfy the joint accuracy–storage criterion.

8.3 Adequate classes with unsuccessful fitted heads

A complementary diagnosis addresses already-examined Bike Sharing decision cohorts (Fanaee-T, 2013): three related targets, 18 fixed maps per target, and 172 observations per cohort. The class consists of 2-Lipschitz functions into $[0, 1]$ under Euclidean distance on stored binary64 features. Original labels outside $[0, 1]$ remain in the losses. These coordinates are treated as exact inputs; native transform error is not enclosed.

Verified primal assignments and weak-dual witnesses bracket the full finite-cohort class optimum. All 54 classes have constructive upper bounds below their unchanged tolerances. In 39 of those classes both inherited fitted heads have risk lower bounds above the tolerance; all 108 fitted heads have strictly positive certified empirical excess over the class optimum. The maximum class-risk bracket width is approximately 2.181×10^{-14} normalized MSE.

Target	Adequate classes	Inadequate classes	Both heads fail	Heads with positive excess
Casual	18	0	3	36
Registered	18	0	18	36
Total	18	0	18	36

Table 4: Finite-cohort class-versus-head diagnosis on Bike Sharing. Total equals casual plus registered; the rows are not independent domain replications. Adequacy witnesses use the assessed labels and are not predictors trained before assessment.

The consequence is decisive for interpreting the earlier failure to screen these classes. If $R^{\star}_{c} \leq U_c \leq \tau_c$, every sound lower bound obeys $L_c \leq \tau_c$ and cannot exclude class c at that tolerance.

A stronger optimizer or pair search cannot repair that particular exclusion objective. This does not identify the cause of each fitted head's gap: narrower head families, training, regularization, sample limitations and assessment shift remain possible. It also does not show that a future-accurate decoder can be learned from the available training data.

8.4 Predictive and operational limits

The operational studies assess whether certification reduces computation or deployment cost. A prospective screening study on Beijing air-quality data (Chen, 2015) yielded no class exclusion among 18 candidates and added 18.899588 seconds. Its three-target follow-up avoided zero additional fits. The deployment comparison uses five replications of one HMM law and one chronological Appliances record (Candanedo, 2017). Its two KAHM procedures each met the observed accuracy allowance in six of six cases but achieved byte saving and latency criteria in none, giving zero joint utility passes. These cases establish neither screening savings nor a deployment advantage.

Follow-ups with full-anchor Lipschitz heads and certified optimization resolution did not consistently close strong-comparator gaps. Some improved an inherited head substantially while still missing the prespecified external baseline requirement. Candidates resolved during optimization analysis had already been fitted; their elimination is not a count of saved fits. The two-domain affine-interface comparison did not meet its primary criterion. An affine ablation on those exposed periods provides diagnostic evidence only.

A frozen affine forecasting study assesses three newly examined periods in known domains. A raw-feature primary policy meets all six prespecified finite-cohort inequalities requiring at most 5% greater MSE than a conventional pool and its ridge subset (Appendix C). All 270 primary numerical packets are accepted. The narrowest accuracy margin is 0.0228 percentage points; some monthly comparisons exceed the allowance. This is a positive result for that fixed policy and cohort criterion, not a statistical noninferiority or population-risk result. KAHM's Jena MSE remains 31.82% above ridge in the primary-seed secondary panel.

The practical lesson is that certificates can change the justified diagnosis without changing a forecast or reducing its cost. Their value must be measured against the specific decision to be supported, the class actually certified, and the full cost of obtaining the evidence.

9 Discussion and limitations

Significance for representation learning. The central object is the attainable risk of a declared representation–decoder pair. A risk lower certificate supports a conclusion that a fitted-model comparison cannot: every member fails the specified requirement. A constructive class upper witness can support the opposite diagnosis even when evaluated fits fail. The centered hierarchy supplies increasingly complete population lower information; the empirical studies show when that information changes a justified conclusion and when it cannot.

What the hierarchy does not identify. The theorem concerns bounded scalar targets, a fixed representation metric, and a fixed sensitivity budget. Compact support and unconditional iid sampling enter the consistency statement. A common-origin conditional construction can converge to an average of conditional optima instead of the optimum of one shared decoder. Vector-output and arbitrary dependent-series consistency do not follow. A positive lower endpoint can arise from irreducible uncertainty or decoder restrictions as well as representation information loss.

Three distinct sources of inconclusiveness. A finite observation block can relax too many cross-observation constraints; a valid statistical endpoint can lack power; and a numerical lower witness can be loose. Increasing the optimization effort cannot remove an impossible calibration floor. Increasing block size can reduce population bias while worsening statistical resolution. When a class already has a constructive adequate witness on a cohort, no valid class-exclusion lower bound can cross the same threshold.

Scope of representation and cost claims. Class nesting preserves feasible functions, not the quality of a particular regularized fit. Exact reparameterization can preserve information while changing a Euclidean Lipschitz class or the ease of learning within it. The selected KAHM preprocessing and caps remain task dependent. Strong raw, kernel and conventional predictors prevent the certificate construction from being mistaken for a general accuracy advantage. Minimum feasible cost is relative to a declared family and cost functional; feature count alone does not certify end-to-end latency or storage.

Sampling and adaptation. Independent calibration units and the declared confidence ledgers support the synthetic studies' risk statements; repeated learning runs describe procedure variability. A deterministic census and chronological benchmark periods do not inherit those population guarantees. Nearby windows, fitting seeds on the same targets, and algebraically related responses are not independent population replications. Development selection must precede independent calibration, or its additional adaptivity must be covered. Reusing exposed cohorts is valuable for diagnosis but remains labeled as such. No campaign-wide significance claim is obtained by counting completed studies.

Numerical assurance. Weak duality and constructive feasibility can be checked independently of a numerical solver's termination status. Exact arithmetic still depends on the chosen function semantics and trusted inputs. Native feature enclosures demonstrated on bounded manufactured configurations do not automatically certify every feature computation in the public-data studies. The later affine output checker assumes trusted stored coordinates, recomputes an exact reference, and has measurable overhead. It provides neither cryptographic authentication nor a speedup over direct exact recomputation.

Further applications. A public-data population application needs a defensible sampling or dependence model and an admissible predictor under the same loss and class. Certificate-directed savings from avoiding fits, reliable adaptive discovery of refinements, and end-to-end native feature computation certification remain separate extensions. They are not premises of the scoped results presented here.

10 Conclusion

Prediction-risk certification should distinguish the performance of a fitted decoder from the best risk allowed by its representation and sensitivity class. The centered scalar hierarchy connects a pairwise obstruction to that optimum under explicit conditions. Valid calibration, verified witnesses, and fitted-risk upper bounds support interpretable exclusion, adequacy, and family-relative cost decisions. The studies demonstrate both informative certificates and limits on their practical use. Separating population risk, finite-cohort capacity, and numerical output fidelity makes those conclusions usable without overstating generality.

Data and code availability

The manuscript sources, original scientific code, experimental protocols, and numerical records are available in the companion repository. Experiments are organized by scientific question. The supplied scripts reproduce the displayed numerical evidence from frozen inputs, forecasts, and optimization witnesses. Study-specific instructions state the input assumptions, data provenance, and scope of these replays; full regeneration of fitted models from raw data is a separate task.

Use of AI tools. AI tools assisted with derivation exploration, implementation, literature retrieval, auditing, and manuscript preparation.

Appendix A. Why a valid certificate may remain unresolved

This appendix quantifies three distinct limitations: finite-block population bias, calibration uncertainty, and numerical slack. The analysis uses standard covering and bounded-concentration arguments. The bounds are sufficient conditions, rather than optimal sample-complexity rates.

A.1 A finite-cover bound on population block bias

For the scalar class in (1), write $\Delta_b = R^*_\kappa - \mathcal{W}_b \geq 0$. Let $N_{\mathcal{G}}(\eta)$ be the size of a finite internal uniform η-cover on the representation support: every representative belongs to $\mathcal{G}_\kappa(d)$ and every class member is uniformly within η of one representative. The internal condition matters because the representatives must have risks at least R^*_κ.

Proposition 9 (A block-bias bound from an internal cover) *For every $\eta > 0$ admitting such a cover,*

$$0 \leq \Delta_b \leq \min\left\{\frac{B}{4},\ 2D\eta + B\sqrt{\frac{\log N_{\mathcal{G}}(\eta)}{2b}}\right\}. \tag{22}$$

The infimum over admissible η is also a valid upper bound.

Proof Let the internal cover consist of $g_1, \ldots, g_N$. With P_b denoting the empirical average, the squared-loss Lipschitz inequality in the proof of Theorem 2 gives

$$a_b \geq \min_j P_b \ell_{g_j} - 2D\eta \geq R^*_\kappa - \max_j\{R(g_j) - P_b\ell_{g_j}\} - 2D\eta.$$

Each difference in braces is centered and has exponential moment at most $\exp\{t^2B^2/(8b)\}$ by the bounded-variable exponential inequality. Log-sum-exp and Jensen therefore give

$$\mathbb{E}\max_j\{R(g_j) - P_b\ell_{g_j}\} \leq \frac{\log N}{t} + \frac{tB^2}{8b} \qquad (t > 0).$$

Optimizing t yields $B\sqrt{\log N/(2b)}$; when $N = 1$ take $t \downarrow 0$. Since $T_b \geq a_b$ and $\Delta_b \leq R^*_\kappa \leq B/4$, the result follows. ■

For a metric support with a radius-r cover of size $M(r)$ and $\kappa > 0$, take $r = \eta/(4\kappa)$ and divide I into at most $J_\eta = 1 + \lceil 2D/\eta \rceil$ bins of width at most $\eta/2$. Decoders occupying the same bins at all

cover centers differ by at most $2\kappa r + \eta/2 = \eta$ everywhere on the support. Selecting one admissible representative from each occupied pattern proves

$$\log N_{\mathcal{G}}(\eta) \le M\Big(\frac{\eta}{4\kappa}\Big) \log J_\eta. \tag{23}$$

For $\kappa = 0$ the class consists of constants and $\mathcal{W}_b = \mathrm{Var}(Y) = R^*_\kappa$ exactly. The metric entropy can make (22) pessimistic. It controls informativeness of the centered hierarchy; it is not a complexity penalty needed for validity of Proposition 5. Statistical analysis of metric-space Lipschitz regression has direct prior work (Gottlieb et al., 2017).

A.2 Separating the errors that prevent a decision

Use $K = \lfloor n/b \rfloor \ge 1$ predetermined disjoint iid blocks for the coarse class and $N \ge 1$ iid fitted losses for a development-fixed refined head under the same target and evaluation law. Their samples may overlap. Suppose rigorous enclosures satisfy

$$0 \le \underline{T}_k \le T_k \le \overline{T}_k \le C_b, \qquad 0 \le \underline{\ell}_i \le \ell_i \le \overline{\ell}_i \le B.$$

Let $\nu_L = K^{-1}\sum_k(\overline{T}_k - \underline{T}_k)$ and $\nu_U = N^{-1}\sum_i(\overline{\ell}_i - \underline{\ell}_i)$. Form the KL endpoints from the lower block mean and upper loss mean at allocations α_L, α_U. Denote those exact-real endpoints by $L^{\mathrm{KL}}, U^{\mathrm{KL}}$. Suppose the reported values obey

$$L^{\mathrm{KL}} - w_L \le L \le L^{\mathrm{KL}}, \qquad U^{\mathrm{KL}} \le U \le U^{\mathrm{KL}} + w_U, \qquad w_L, w_U \ge 0.$$

These widths include certified KL-root and rescaling errors. Write $r_r = R_r(\widehat{g}_r)$ and define $h(C, m, a) = C\sqrt{\log(1/a)/(2m)}$.

Theorem 10 (A sufficient resolution bound) *The reported risk statements are jointly valid with probability at least $1 - \alpha_L - \alpha_U$. For separately chosen $\beta_L, \beta_U \in (0, 1)$, with probability at least $1 - \beta_L - \beta_U$,*

$$R^*_c - L \le \Delta_b + h(C_b, K, \alpha_L) + h(C_b, K, \beta_L) + \nu_L + w_L, \tag{24}$$
$$U - r_r \le h(B, N, \alpha_U) + h(B, N, \beta_U) + \nu_U + w_U. \tag{25}$$

If E_L and E_U denote the terms after Δ_b in these two bounds, then on that event

$$L - U \ge (R^*_c - r_r) - \Delta_b - E_L - E_U. \tag{26}$$

Validity and resolution hold together with probability at least $1 - \alpha_L - \alpha_U - \beta_L - \beta_U$.

Proof Validity follows from Theorem 6. The binary divergence satisfies $\mathrm{kl}(q\|p) \ge 2(q-p)^2$. For fixed $p \in (0, 1)$, differentiate the difference twice in q: its second derivative is $1/\{q(1-q)\} - 4 \ge 0$, and its value and first derivative vanish at $q = p$. Boundary cases follow by limits. Thus

$$Ck_-(q, c) \ge Cq - C\sqrt{c/2}, \qquad Ck_+(q, c) \le Cq + C\sqrt{c/2}.$$

Write $\bar{T} = K^{-1}\sum_k T_k$ and $\bar{\ell} = N^{-1}\sum_i \ell_i$ for the exact sample means. Hoeffding's lower tail gives $\bar{T} \ge \mathcal{W}_b - h(C_b, K, \beta_L)$ except with probability β_L. Pointwise, $K^{-1}\sum_k \underline{T}_k \ge \bar{T} - \nu_L$. Apply

the lower KL comparison and subtract w_L to obtain (24). Similarly, $\bar{\ell} \le r_r + h(B, N, \beta_U)$ except with probability β_U, and $N^{-1}\sum_i \bar{\ell}_i \le \bar{\ell} + \nu_U$ pointwise. The upper KL comparison and w_U prove (25). Subtraction and union bounds give the remaining conclusions without requiring independence between endpoints. ■

For an a priori power statement, the observed numerical widths must be replaced by deterministic protocol bounds or covered by an additional probability event. Inserting post hoc small widths into an unconditional power promise is not justified. For a fixed MSE tolerance τ, the conditions $R_c^* - \tau > \Delta_b + E_L$ and $\tau - r_r \ge E_U$ are sufficient for coarse-class exclusion and refined-head acceptance. The first is strict because the exclusion decision is strict. The α allocations govern validity; the β allocations govern how often the certificate resolves. A grid of fixed-sample calculations is not an optional-stopping guarantee.

A.3 Safe transfer to empirical Bernstein endpoints

For $N \ge 2$ exact iid scores $Z_i \in [0, C]$, the empirical Bernstein radius is

$$r_{\mathrm{EB}}(Z; C, \alpha) = \sqrt{\frac{2s_Z^2 \log(2/\alpha)}{N}} + \frac{7C\log(2/\alpha)}{3(N-1)}, \qquad s_Z^2 = \frac{1}{N-1}\sum_i (Z_i - \bar{Z})^2.$$

Each one-sided endpoint has failure probability at most α (Maurer and Pontil, 2009, Theorem 4). Lowering individual numerical scores can lower their variance enough to increase a naive lower endpoint; the monotone-mean transfer used for KL is therefore insufficient.

If $[l_i, u_i] \subseteq [0, C]$ encloses each exact score, set $c = (\bar{l} + \bar{u})/2$. A valid variance upper enclosure is

$$\overline{V}_N = \min\left\{ \frac{\sum_i \max\{(l_i - c)^2, (u_i - c)^2\}}{N-1},\ \frac{C^2\lfloor N^2/4 \rfloor}{N(N-1)} \right\}. \tag{27}$$

Indeed, the empirical mean minimizes squared distance to a common center, so $\sum_i (Z_i - \bar{Z})^2 \le \sum_i (Z_i - c)^2$. Every latter summand is bounded by its larger interval-endpoint value. The second quantity is the maximum sample variance on $[0, C]^N$, obtained by placing coordinates at endpoints as in Theorem 2. Both bound the same variance, so their minimum is safe.

Evaluate the radius with a rigorous upper enclosure of (27). Then $\bar{l} - r_{\mathrm{EB}}^{\mathrm{upper}}$ and $\bar{u} + r_{\mathrm{EB}}^{\mathrm{upper}}$, clipped to $[0, C]$, widen the exact-score interval pointwise. Their coverage survives coupled or adaptive numerical computation. With only a verified lower score, the range fallback $[l_i, C]$ is valid but may be weak; the observation must remain included. Alternatively, a fixed local numerical function of each independent block produces iid computed scores that may be calibrated directly. Neither argument validates the sample variance of a data-selected empirical-risk minimizer as a class-risk certificate.

Appendix B. Calibration, finite-cohort witnesses, and study protocols

B.1 Controlled study and numerical scope

Each of the 24 controlled learning repetitions uses disjoint pools of 256 support episodes, 512 head-training episodes, 256 validation episodes, 4096 calibration episodes, and 1024 terminal episodes. The geometry menu crosses min–max or standard scaling, local caps one or two, and concentrations

$0.5, 4, 16, 64$. The two-dimensional input means cap two already permits full ambient rank. Prefix sizes two and six share the geometry selected by the refined prefix; ridge parameters $0, 10^{-4}, 10^{-2}$ are selected separately for the two heads. There is no postselection refit.

The $\kappa = 2$ block-norm constraint and clipping to $[-0.85, 0.85]$ define admissibility. Calibration uses nested prefixes $n = 256, 1024, 4096$ and observation-block sizes $2, 4, 8, 16, 32$. All candidate families and allocations are fixed before calibration. The 1008 one-sided events are $24 \times 2 \times 3 \times (5 + 1 + 1)$: five block lower endpoints, one recycled lower endpoint, and one fitted upper endpoint. Equal allocation per endpoint is not equal aggregate allocation between method families.

Both calibration methods share 190464 block programs over the full 4096-observation workload. Reuse adds KL inversion and bookkeeping, not a second solve. The independent audit regenerates sampled observations, checks candidate objectives and selection, replays dual values, and recomputes endpoints. Common native feature sentinels and guarded floating arithmetic do not constitute an independent end-to-end interval proof. The mathematical confidence statement assumes numerically valid inputs and witnesses; the computational evidence retains that limitation.

B.2 Finite-cohort risk is a deterministic optimization problem

Fix coordinates $z_1, \dots, z_M$, targets $y_1, \dots, y_M$, a metric d, and output interval $I = [a, b]$. Targets in this subsection need not belong to I. For

$$\mathcal{F} = \{f : \mathcal{U} \to I : \ |f(u) - f(v)| \le \kappa d(u, v)\}, \qquad R_M^\star = \inf_{f \in \mathcal{F}} M^{-1} \sum_i (f(z_i) - y_i)^2,$$

scalar extension gives

$$R_M^\star = \frac{1}{M} \min_{v \in I^M} \left\{ \sum_i (v_i - y_i)^2 : |v_i - v_j| \le \kappa d(z_i, z_j) \quad (i < j) \right\}. \tag{28}$$

Restriction of any class member is feasible. Conversely, a feasible vector is interpolated by $\mathrm{clip}_I \min_i\{v_i + \kappa d(\,\cdot\,, z_i)\}$. A finite minimum of equally Lipschitz scalar functions is Lipschitz, and the pair inequalities imply interpolation, including duplicate coordinates. Clipping is nonexpansive. The feasible set is nonempty and compact, so its minimum exists. This is classical Lipschitz extension and constrained regression (Gottlieb et al., 2017).

Let B_{e} be an oriented edge-incidence matrix with rows $e_i - e_j$. Given rigorous distance upper bounds d_{ij}^+ and any signed rational edge vector λ, put $s = B_{\mathrm{e}}^{\mathsf{T}} \lambda$. Weak duality gives the lower witness

$$D(\lambda) = \sum_i \min_{t \in I} \{(t - y_i)^2 + s_i t\} - \sum_{i<j} \kappa d_{ij}^+ |\lambda_{ij}| \ \le \ M R_M^\star. \tag{29}$$

Indeed, $\lambda^{\mathsf{T}} B_{\mathrm{e}} v \le \sum \kappa d_{ij}^+ |\lambda_{ij}|$ for every feasible v, and minimization over the larger box preserves the lower inequality. The scalar minima occur at $\mathrm{clip}_I(y_i - s_i/2)$. A rigorously feasible vector supplies an upper witness. With exact arithmetic one may take $L = \max\{0, D(\lambda)/M\}$ and $U = M^{-1} \sum_i (v_i - y_i)^2$ for such a vector; conservative outward enclosures preserve these directions. Consequently $L \le R_M^\star \le U$ can be verified without treating solver termination as proof.

With a fitted member h and a verified cohort-risk interval $[\ell_h, u_h]$,

$$\max\{0, \ell_h - U\} \le R_M(h) - R_M^\star \le u_h - L.$$

For an MSE tolerance τ, class inadequacy requires $L > \tau$, while constructive adequacy requires $U \leq \tau$. A fitted head is proved to fail only when $\ell_h > \tau$; an upper endpoint above the threshold is insufficient. These statements use every fixed observation and require no sampling penalty. They do not imply a population-risk statement or a method for learning the witnessed function before seeing the cohort.

B.3 Natural cohort and class-versus-head diagnosis

The PTB-XL census reuses 4096 calibration and 4096 terminal rows from an already examined study. No new patient cohort was opened. The two-, six-, and fourteen-coordinate maps were selected separately; their relation is not feature nesting. Consecutive and feature-neighbor partitions of sizes 32, 64, and 128 relax the full-cohort constraints, with exact selected dual witnesses. In this census calculation, feature-neighbor grouping is a deterministic optimization device and is not asserted to be an iid pairing scheme. Canonical-feature enclosures are inherited premises. The source is PTB-XL version 1.0.3 (Wagner et al., 2020, 2022).

The Bike Sharing diagnosis uses the same previously exposed 172 decision observations per target and 18 maps (two raw, four Nyström, twelve KAHM). It retains both scalings, KAHM caps $4, 12, 20$, two prefix choices, and original target values. There are 54 target/map classes and 108 inherited heads. Rational distance endpoints use 80-bit dyadic square-root floors checked by exact squared inequalities. Numerical QP proposals are repaired into feasible vectors using shortest-path Lipschitz envelopes and an exact convex blend. Exact dual values and the common range floor provide the lower endpoints. All 54 class brackets satisfy the prespecified precision flag; no numerical fallback occurs.

All 54 classes also have an adequate target-envelope witness without using the solver proposal, although the solver tightens the brackets. This diagnostic uses assessment labels and therefore is not a training-only repair. The three responses are casual, registered, and their sum; the data do not supply three independent applications (Fanaee-T, 2013).

B.4 Predictive and operational comparisons

The accompanying records preserve complete configurations and outcomes for the following comparisons.

- **Deployment cost.** Five replications of one HMM law and one chronological Appliances record are assessed. Each of two KAHM procedures has six accuracy passes, zero byte-saving passes, zero latency passes, and zero joint deployment-utility passes.

- **Public-data screening.** The Beijing and Bike Sharing panels yield no useful certificate-specific screening and no additional fits avoided. The Bike Sharing class-versus-head analysis establishes why exclusion is impossible for the same adequate classes and tolerances.

- **Decoder diagnostics.** Stronger decoder families, regularization and extension choices, and optimization-resolution checks give mixed predictive results. Seoul (UCI Machine Learning Repository, 2020) misses the strong comparator despite a large improvement over its inherited head. The Tetouan/Metro joint selector (Salam and El Hibaoui, 2018; Hogue, 2019) retains the default and fails both tasks; the Household/Jena comparison passes one task and fails the other.

- **Affine-interface diagnostics.** The two-domain primary forecasting criterion is not met. The subsequent affine ablation reuses exposed periods and includes a strong raw-feature control and cases where norm projection or clipping leaves the selected predictions unchanged.

These protocols have different targets and evaluation scopes, so their outcomes are not pooled into one hypothesis test. Secondary and diagnostic comparisons do not replace failed primary criteria.

Appendix C. Supporting study: conditional numerical output fidelity

The frozen affine forecasting study evaluates whether a numerically checked affine interface can meet a declared finite-cohort accuracy allowance under strong baseline controls. It is separate from population class-risk certification.

C.1 Frozen policy and assessed periods

At each daily 23:00 forecast origin, the task predicts the same variable 24 hours ahead using the 24 hourly snapshots available at or before that origin. Household uses minute-00 minute-average power, Jena minute-00 ten-minute temperature observations, and Metro reported hourly traffic volume. Household September–November 2008 is a newly assessed historical holdout within the retained workflow; Jena September–November 2018 and Metro July–September 2018 are forward periods. Each has 90 eligible queries. These are new assessed periods in known domains, not evidence of unseen-domain generalization or universal historical non-exposure.

Three chronological development folds precede final fitting. The candidate factory includes two raw maps, four Nyström maps (widths 8 and 24), and twelve KAHM maps (caps $4, 12, 20$ and prefixes one and two), crossing min–max and standard scaling. Five mean-loss ridge penalties, $10^{-6}, 10^{-4}, 10^{-2}, 1, 100$, give 90 affine candidates. A training-derived metric scalar remains fixed. Coefficients are contracted if needed to Euclidean norm at most two, the intercept is recentered exactly, and normalized output is clipped to $[0, 1]$.

Selection minimizes pooled exact physical-unit development MSE with fixed lexical ties. Seed 3901 is primary; seeds 4301 and 4302 provide descriptive fitting sensitivities. The conventional reference pool has 44 candidates; its five-candidate ridge subset is a second reference. Only selected final candidates are assessed, with every policy/seed forecast alias sealed before its target is released.

Task (units)	Raw RMSE	Pool RMSE	Ridge RMSE	Δ vs pool	Δ vs ridge
Household (kW)	1.0969	1.0848	1.0709	$+2.24\%$	$+4.91\%$
Jena (°C)	2.6062	2.5437	2.5437	$+4.98\%$	$+4.98\%$
Metro (vehicles/h)	519.2678	531.2858	523.9567	-4.47%	-1.78%

Table 5: Primary raw-feature policy on the three assessment periods. Percentage changes are in MSE, not RMSE. The two Jena references select the same ridge model.

The primary criterion is the conjunction of six exact inequalities $20\,\mathrm{SSE}_{\mathrm{raw}} \le 21\,\mathrm{SSE}_{\mathrm{reference}}$, acceptance of every primary packet, and the prescribed independent audit. All pass. Household is 4.9120% above ridge MSE and Jena is 4.9772% above its coincident references, leaving margins of 0.0880 and 0.0228 percentage points. The result is deterministic for these cohorts, not a statistical noninferiority test.

At the primary seed the all-representation selector chooses KAHM, raw, and Nyström on Household, Jena, and Metro. Its descriptive pass does not replace the primary raw policy. KAHM RMSE is 1.0828, 2.9204, and 530.6287 respectively; its Jena MSE is 31.82% above ridge. Raw and Nyström predictions are unchanged across the three seeds, which reuse the same targets.

C.2 Exact reference and checker contract

Trusted stored binary64 feature coordinates z, a fixed rational affine head, and training target offset a and positive span s define

$$F(z) = a + s\ \mathrm{clip}_{[0,1]}(b + \beta^{\mathsf{T}} z), \qquad \|\beta\|_2 \le 2.$$

The reference lies in $[a, a+s]$ and is at most $2s$-Lipschitz in the supplied coordinates. The unchanged packet producer and independent checker bind the expected deployment, query, feature bytes, and reported output. Integer arithmetic evaluates the exact reference and outward numerical bounds. Acceptance requires

$$|\widehat{y} - F(z)| \le s\, 2^{-40}.$$

The reported binary64 value may lie outside an exact range endpoint by up to this budget. The claim concerns the stored head and supplied coordinates. It does not certify an exact training optimum, ideal feature-transform arithmetic, raw-input robustness, authentication, or coverage of a future label.

Across 15 distinct selected affine deployments and 90 queries each, all 1350 honest outputs are accepted. Every tested float32 conversion changes the reported value; 1349 are over budget and rejected, while the one in-budget case is accepted. All 45 prescribed wrong-query, wrong-deployment, and stale-feature challenges are rejected. A deliberately limited binding-consistency control accepts all self-consistent float32 cases; it is not a comprehensive security baseline.

Task	Plain inference	Full certified	Cached checker	Cached exact
Household	0.018224	0.244745	0.061065	0.017733
Jena	0.018218	0.240890	0.061034	0.017559
Metro	0.018190	0.245817	0.060262	0.017273

Table 6: Primary raw-policy costs in milliseconds: median across 32 prescribed queries of the per-query median over ten repeats after three warmups. Full inference includes the map and head; cached comparisons use trusted features and static context. Local I/O, process startup and network cost are outside these primitive timings.

Full certified inference is approximately 13.2–13.5 times plain inference on this panel. The cached checker is slower than direct exact evaluation of the affine reference, which it recomputes. This implementation is neither a succinct proof system nor a verification speedup. Sampled packet sizes are 447–453 bytes, separate from feature and static-context storage. The tiny numerical tolerance has not been linked to an externally motivated application decision, and detecting a precision conversion alone does not establish harmful forecast error.

C.3 Limits and reproducibility

All months remain in the aggregate comparison. September raw MSE exceeds ridge by 10.28% on Household and 12.53% on Metro; Jena exceeds its reference by 8.78%. Monthly results are descriptive. The 270 dated targets are dependent observations from three known domains; no population guarantee follows from their aggregate pass.

The retained execution contains 27 producer commands and three independent audit phases without outcome-based reruns. The reproduction records include exact losses, numerical packets, event logs, timings, and selected configurations. A standalone demonstration reproduces three honest acceptances and three expected float32 rejections on the fixed first queries. Hashes establish artifact consistency within retained records, not external attestation.

Data come from the UCI Household power and Metro traffic datasets (Hebrail and Berard, 2006; Hogue, 2019), and the MPI-BGC Beutenberg weather archives (Max Planck Institute for Biogeochemistry, n.d.). Source URLs and hashes are retained in the original package. Official-archive byte identity of the pinned Household mirror remains unverified. Eligibility metadata was accessed before the measurement freeze as declared. These details bound the freshness and provenance claims.

Earlier manufactured-configuration experiments supplied native feature-evaluator enclosures under their own assumptions. The affine study's use of exact stored coordinates does not inherit such enclosures for its public-data feature pipeline. Certifying that additional computation would be a separate extension; it is not needed for the conditional numerical contract reported here.

References

Aditya Kumar Akash, Vishnu Suresh Lokhande, Sathya N. Ravi, and Vikas Singh. Learning invariant representations using inverse contrastive loss. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(8):6582–6591, 2021. doi: 10.1609/aaai.v35i8.16815. URL https://ojs.aaai.org/index.php/AAAI/article/view/16815.

Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. Learn then test: Calibrating predictive algorithms to achieve risk control. *The Annals of Applied Statistics*, 19(2):1641–1662, 2025. doi: 10.1214/24-AOAS1998. URL https://arxiv.org/abs/2110.01052.

Francis R. Bach. Consistency of the group lasso and multiple kernel learning. *Journal of Machine Learning Research*, 9(40):1179–1225, 2008. URL https://jmlr.org/papers/v9/bach08b.html.

Luis Candanedo. Appliances energy prediction. UCI Machine Learning Repository, 2017. URL https://archive.ics.uci.edu/dataset/374/appliances+energy+prediction. Dataset 374.

Song Chen. Beijing PM2.5. UCI Machine Learning Repository, 2015. URL https://archive.ics.uci.edu/dataset/381/beijing+pm2+5+data. Dataset 381.

Hadi Fanaee-T. Bike sharing. UCI Machine Learning Repository, 2013. URL https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset. Dataset 275.

Lee-Ad Gottlieb, Aryeh Kontorovich, and Robert Krauthgamer. Efficient regression in metric spaces via approximate lipschitz extension. *IEEE Transactions on Information Theory*, 63(8):4838–4849, 2017. doi: 10.1109/TIT.2017.2713820.

Vincent Guigues, Anatoli Juditsky, and Arkadi Nemirovski. Non-asymptotic confidence bounds for the optimal value of a stochastic program. *Optimization Methods and Software*, 32(5):1033–1058, 2017. doi: 10.1080/10556788.2017.1350177. URL https://www2.isye.gatech.edu/~nemirovs/GuiJudNem2017.pdf.

Georges Hebrail and Alice Berard. Individual household electric power consumption. UCI Machine Learning Repository, 2006. URL https://archive.ics.uci.edu/dataset/235/individual+household+electric+power+consumption. Dataset 235.

Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830. URL https://www.tandfonline.com/doi/abs/10.1080/01621459.1963.10500830.

John Hogue. Metro interstate traffic volume. UCI Machine Learning Repository, 2019. URL https://archive.ics.uci.edu/dataset/492/metro+interstate+traffic+volume. Dataset 492.

Nan Jiang, Alex Kulesza, and Satinder Singh. Abstraction selection in model-based reinforcement learning. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 179–188, 2015. URL https://proceedings.mlr.press/v37/jiang15.html.

Mohit Kumar and Somayeh Kargaran. Prediction limits and Koopman closure of geometry-induced soft state abstractions. arXiv:2609.32652v3, 2026. URL https://arxiv.org/abs/2609.32652v3.

Mohit Kumar, Bernhard A. Moser, and Lukas Fischer. On mitigating the utility-loss in differentially private learning: A new perspective by a geometrically inspired kernel approach. *Journal of Artificial Intelligence Research*, 79:515–567, 2024. doi: 10.1613/jair.1.15071. URL https://jair.org/index.php/jair/article/view/15071.

Mohit Kumar, Alexander Valentinitsch, Magdalena Fuchs, Mathias Brucker, Juliana Bowles, Adnan Husakovic, Ali Abbas, and Bernhard A. Moser. Geometrically inspired kernel machines for collaborative learning beyond gradient descent. *Journal of Artificial Intelligence Research*, 83 (16), 2025. doi: 10.1613/jair.1.16821. URL https://jair.org/index.php/jair/article/view/16821.

Jeff Linderoth, Alexander Shapiro, and Stephen Wright. The empirical behavior of sampling methods for stochastic programming. *Annals of Operations Research*, 142(1):215–241, 2006. doi: 10.1007/s10479-006-6169-8. URL https://jlinderoth.github.io/papers/Linderoth-Shapiro-Wright-06.pdf.

Wai-Kei Mak, David P. Morton, and R. Kevin Wood. Monte Carlo bounding techniques for determining solution quality in stochastic programs. *Operations Research Letters*, 24(1–2):47–56,

1999. doi: 10.1016/S0167-6377(98)00054-6. URL https://www.sciencedirect.com/science/article/pii/S0167637798000546.

Andreas Maurer and Massimiliano Pontil. Empirical Bernstein bounds and sample variance penalization. In *Proceedings of the 22nd Annual Conference on Learning Theory*, 2009. URL https://arxiv.org/abs/0907.3740.

Max Planck Institute for Biogeochemistry. Weather data, n.d. URL https://weather.bgc-jena.mpg.de/weather_data.html. Beutenberg station archives. Accessed 8 November 2026.

Manuel M. Müller, Yuetian Luo, and Rina Foygel Barber. Are all models wrong? fundamental limits in distribution-free empirical model falsification. In *Proceedings of the 38th Conference on Learning Theory*, volume 291 of *Proceedings of Machine Learning Research*, pages 4271–4308, 2025. URL https://proceedings.mlr.press/v291/muller25a.html.

Abdulwahed Salam and Abdelaaziz El Hibaoui. Power consumption of tetouan city. UCI Machine Learning Repository, 2018. URL https://archive.ics.uci.edu/dataset/849/power+consumption+of+tetouan+city. Dataset 849.

UCI Machine Learning Repository. Seoul bike sharing demand, 2020. URL https://archive.ics.uci.edu/dataset/560/seoul+bike+sharing+demand. Dataset 560.

Patrick Wagner, Nils Strodthoff, Ralf-Dieter Bousseljot, Dieter Kreiseler, Fatima I. Lunze, Wojciech Samek, and Tobias Schaeffter. PTB-XL, a large publicly available electrocardiography dataset. *Scientific Data*, 7:154, 2020. doi: 10.1038/s41597-020-0495-6. URL https://www.nature.com/articles/s41597-020-0495-6.

Patrick Wagner, Nils Strodthoff, Ralf-Dieter Bousseljot, Wojciech Samek, and Tobias Schaeffter. PTB-XL, a large publicly available electrocardiography dataset. PhysioNet, 2022. URL https://physionet.org/content/ptb-xl/1.0.3/. Version 1.0.3.